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ORIGINAL RESEARCH

Role of Hip Internal Rotation Range and Foot Progression Angle for Preventing Jones Fracture During Crossover Cutting

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Jones fracture is a common injury in soccer players, caused by overload on the fifth metatarsal (M5) that can occur during crossover cutting. This study aimed to investigate the effects of the hip internal rotation (HIR) range in passive and dynamic conditions and foot progression angle (FPA) on the forces applied to the M5 during crossover cutting. The study included 20 men with experience playing soccer. The passive HIR was measured in the prone position. A 3-dimensional motion analysis system, force plates, and Footscan were used to measure the angle of the lower limb, including dynamic HIR, FPA, and plantar pressure to the M5, during crossover cutting in the 45° direction. Pearson correlation coefficient was measured to examine the relationship between variables. Passive and dynamic HIR were not related to the plantar pressure on the M5 ($P > .05$). However, increased FPA in movement direction correlated with a reduction in plantar pressure on the M5 ($r = -.56$, $P < .01$). Minimizing the FPA with respect to the new direction during crossover cutting reduced plantar pressure on the M5 and thus could prevent Jones fracture.

Keywords: fifth metatarsal, plantar pressure, stress fracture, soccer

Jones fracture is a stress fracture of the proximal fifth metatarsal (M5) that frequently occurs in male soccer players. According to a stress fracture epidemiology study published by Ekstrand and Torstveit,[1] Jones fracture accounted for 0.5% of all injuries and 78% of all stress fractures in male soccer players. It is intractable,[2] leading to prolonged healing and nonunion due to the damage of nutrient vessels in the proximal M5.[3] After a Jones fracture, surgical treatment is recommended because it results in a higher return-to-competition rate than conservative treatment.[4] It takes approximately 11 weeks to return to competition after surgery, the refracture rate is 22.2% to 25.0%,[1,5] and the chance to participate in games is lower in the year after returning to competition than before injury.[5] Therefore, it is important to prevent a Jones fracture in soccer players.

Jones fracture occurs due to an increased bending moment to the proximal portion of M5. The bending moment is defined as the product of the force magnitude on the M5 and the distance from the point of force application to the proximal M5.[6] The proximal M5 is tightly fixed by muscles and ligaments, such as the peroneus brevis muscle tendon, peroneus tertius muscle tendon, and lateral plantar fascia,[7] and the bending moment might be concentrated at the proximal M5 when a large force is applied to the M5 head.[6] However, it is difficult to measure the bending moment during competition in vivo. A recent study revealed that patients with a history of Jones fracture have increased plantar pressure on the M5 during gait compared with players without a history of Jones fracture.[8] Therefore, increased plantar pressure on the M5 is related to a large bending moment, which might be involved in the Jones fracture occurrence.

Crossover cutting, one of the competitive movements in soccer, increases plantar pressure on the lateral surface of the foot more than other movements in soccer.[9] Crossover cutting is a sudden change of direction that is performed by landing the foot on the same side as the new direction and then crossing the opposite leg in front to provide acceleration in the new direction.[10] As the movement direction is on the same side as the landing foot, plantar pressure to the lateral surface of the foot is high. Considering that the change of direction is repeated more than 600 times during a soccer game[11] and stress fractures are caused by repetitive and excessive loading on the bones,[12] it is necessary to examine the factors that increase plantar pressure on the M5 during crossover cutting. Nevertheless, it remains unclear which cutting technique is related to the plantar pressure on the M5.

The hip internal rotation (HIR) range and foot progression angle (FPA) may influence plantar pressure on the M5 during crossover cutting. In soccer players, the limitation of passive HIR is a risk factor for developing Jones fracture.[13] Crossover cutting involves pelvic horizontal rotation toward the supporting leg.[10] Regarding closed kinetic chain, pelvic horizontal rotation with HIR limitation may result in compensatory femoral and lower leg external rotation (ER). As lower leg ER is associated with foot supination,[14] it increases plantar pressure on the M5. Furthermore, FPA, which indicates the direction of the foot during movement, alters foot kinematics and the load on the plantar surface.[15,16] FPA is the angle between the longitudinal axis of the foot and the forward line of progression.[17] Previous studies concerning walking and running have reported that a great FPA increases subtalar pronation[15] and reduces plantar pressure on the lateral plantar surface of the foot.[16] Moreover, those with flat feet who exhibit overpronation of the foot have less plantar pressure on the lateral plantar surface of the foot compared with those with normal feet.[18] This observation means that an increase in FPA increases subtalar pronation, decreasing the plantar pressure on the M5. However, the

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effects of passive and dynamic HIR and FPA on the plantar pressure acting on the M5 during crossover cutting have yet to be clarified.

This study aimed to examine the association of passive and dynamic HIR and FPA with plantar pressure on the M5 during crossover cutting. We hypothesized that a greater passive HIR is associated with a greater dynamic HIR and a smaller plantar pressure on the M5 and that a greater FPA leads to a smaller plantar pressure on the M5. This study's findings could help develop strategies for improving the management and treatment of Jones fracture.

Methods

Participants

This study included 20 male college students with soccer experience who competed at the regional league level. The mean values (mean [SD]) of the subjects' ages, heights, weights, and years of soccer experience were 21.0 years (0.8), 169.9 cm (4.7), 61.6 kg (7.8), and 10.3 years (2.4), respectively. Participants had no previous musculoskeletal injury to the lower extremity within the past 6 months, no previous injury to the lower extremity requiring surgery (eg, fracture), and ≥3 years of experience playing soccer. This study was approved by the ethics committee of Niigata University of Health and Welfare (No. 18680-210722). All participants provided written informed consent to participate in the study.

Equipment

The study used a 3-dimensional motion analysis system (VICON), including 12 infrared cameras and 2 force plates (OR6-6-2000, AMTI). The sampling frequencies were 250 and 1000 Hz. Footscan interface box systems (Footscan, Rsscan) were used to measure plantar pressure. It consists of a pressure plate (1068 mm × 418 mm × 12 mm) and an interface box. The pressure plate had 8192 resistive sensors arranged in a 128 × 64 matrix and a 975 mm × 325 mm active surface area. The sensors had the ability to measure plantar pressure with a range of 1 to 127 N/cm^2. The data acquisition frequency was 250 Hz, and the resolution was 12 bits. The plate was connected to the computer. The interface box used the supplied cable, and it was synchronized with the 3-dimensional motion analysis system during measurement through the interface box. A total of 28 reflective markers (14 mm in diameter) were placed on the body surface. Segments are defined for the (1) pelvis: right and left anterior superior iliac spine (ASIS) and right and left posterior superior iliac spine; (2) thigh: right and left greater trochanter, right and left lateral thigh surface, right and left medial epicondyle, and right and left lateral epicondyle; (3) lower leg: right and left fibular head, right and left anterior tibia crest, right and left medial malleoli, and right and left lateral malleoli; and (4) foot: right and left calcaneal ridge, right and left first metatarsal head (MH1), right and left M5 head (MH5), and right and left base of M5. For the segment definition, the origin, first axis, and second axis were defined, and the third axis was determined using the right-hand screw rule. The details of each segment definition are described (origin, first axis, second axis, and token alternative) as follows: pelvis (midpoint between the right and left ASIS, right ASIS–left ASIS, right superior iliac spine–right ASIS, xzy), thigh (knee joint center, hip joint center–knee joint center, medial epicondyle–lateral epicondyle, zyx), lower leg (ankle joint center, knee joint center–ankle joint center, medial malleolus–lateral malleolus, zyx), and foot (midpoint between MH1 and MH5, midpoint between MH1 and MH5-calcaneal ridge, MH5-midpoint between MH1 and MH5, yzx). The ankle joint center was defined as the midpoint of the medial and lateral malleolus, and the knee joint center was defined as the midpoint of the medial and lateral epicondyles. The hip joint center was calculated based on the distance between the left and right ASIS.[19]

Experimental Protocol

Passive HIR was measured by a physical therapist when the subject was positioned prone on the examination table with the hip joint at 0° extension and the knee joint at 90° flexion.[13,20] The intraclass correlation coefficients (1, 3) of the passive HIR was as high as .98, almost perfect according to the criteria reported by Landis and Koch[21] A static standing trial was performed to define the segment axes before the movement task was performed. The movement task was barefoot crossover cutting, and the angle of change in direction was 45°, which is frequently performed during soccer games[11,22] (Figure 1). Specifically, (1) the starting limb position was with the legs shoulder-width apart and arms folded in front of the chest, and a run-up was performed from the starting position toward the Footscan (Figure 1A and 1B). In this study, the upper arm was placed in front of the chest to minimize factors affecting the kinematics of lower limb joints. The distance from the starting position to the Footscan was set to 70% of the maximum horizontal jump distance[23,24] to control the task conditions for each subject (Figure 2). For the horizontal jump, participants jumped as far forward as possible from a standing position on both legs and landed on their nondominant legs (contralateral side of the leg used for kicking when accuracy is required). The distance from the standing position to the heel of the landing foot was measured.[23,24] The horizontal jump was measured 3 times, and the maximum value among the 3 trials was considered as the maximum jump

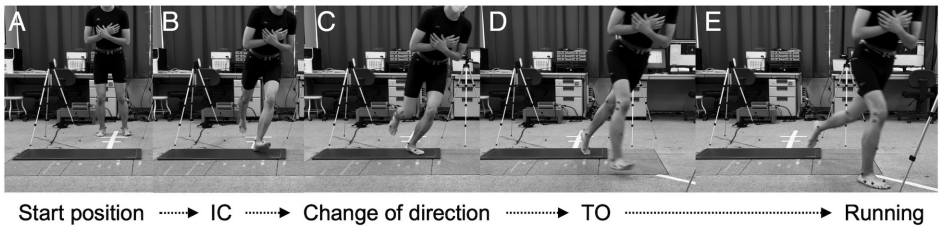

Figure 1 — Crossover cutting. Crossover cutting includes (1) run-up (A, B), (2) landing on the Footscan with the nondominant leg (the opposite side of the leg used for kicking when accuracy was required; B), (3) change in direction by swinging the contralateral leg to the supporting leg (nondominant leg; C), and (4) TO and running in the direction of movement (D, E). Participants performed the task with both upper limbs in front of their chest from the beginning to the end. All participants performed the movement task with rearfoot contact and maximum effort. IC indicates initial contact; TO, toe-off.

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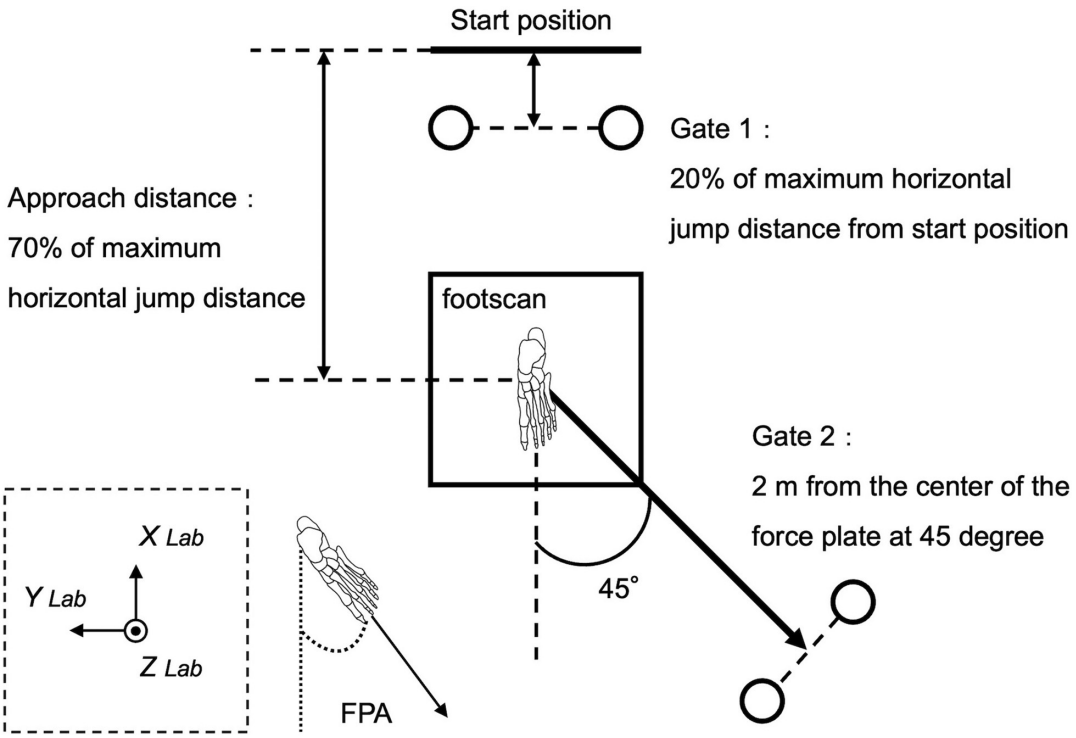

Figure 2 — Overview of the movement task. Gates 1 and 2 show light gate sensors, which were used to measure the movement speed. Measurements began when participants entered gate 1 and ended when they passed gate 2. The velocity (in meters per second) from gates 1 to 2 was used as the result. FPA indicates foot progression angle.

distance. (2) After the run-up, participants stepped onto the Footscan in 1 step using their nondominant legs (Figure 1B). Of the 20 participants, 19 cut to the left, whereas 1 participant cut to the right. The FPA was then adjusted freely by each participant. (3) After foot contact, participants were asked to cross and swing the contralateral leg (dominant leg) against the supporting leg (nondominant leg; Figure 1C), and (4) run 2 m to the new direction of movement (Figure 1D and 1E). Each participant performed the task with maximum effort, and 5 trials of the movement task were measured. As differences in the performance speed among participants in each trial may cause errors in the kinematic variables, the performance speed in the movement task was measured. Light gate sensors (Dashr, S&C Inc) were placed at 20% of the maximum jump distance from the start position (gate 1) and at 2 m away from the foot contact position (gate 2; Figure 2), and the velocity from gates 1 to 2 was calculated as the speed (in meters per second) to perform the movement task.[23,24] The height of the light gate sensor was adjusted to the pelvis height of the subject. Experiments were continued until 5 trials were included in the analysis, excluding trials with an error of more than 0.5 m/s within subjects. For the movement task, the mean speed (mean [SD]) was 4.72 m/s (0.38), and the intertrial speed error in 5 trials was 0.28 m/s (0.15).

Data Analysis

Marker trajectory data were filtered using a 0-lag fourth-order Butterworth filter with a 15-Hz cutoff frequency using Vicon Nexus software (version 2.8.1, VICON).[23,24] BodyBuilder (version 2.8.1; VICON) was used to calculate the HIR angle, thigh ER angle, lower leg ER, ankle eversion angle, and FPA of the supporting leg during the movement. In this study, the 3-dimensional joint angles were calculated at the distal segment and expressed relative to the adjacent proximal segment using right-handed orthogonal Euler xyz rotation. The ER angles of the thigh and lower leg were defined as the rotation angle of the local coordinate system around the *z*-axis of the absolute coordinate system. The absolute coordinate system was defined as the *y*-axis for the right direction from the starting position of the task movement, the *x*-axis for the backward direction, and the *z*-axis for the vertical direction from the floor (Figure 2). The FPA was defined as the angle between the forward line of progression at run-up and the longitudinal axis of the foot (Figure 2). Data obtained by Footscan were analyzed using Footscan 9 software (Footscan, Rsscan). The foot strike pattern of each subject was identified based on the position of the center of pressure at initial contact (IC) as measured by Footscan (Figure 3A) and its location within the foot region classified by the software. The foot was automatically classified by the software into the following zones: medial heel, lateral heel, midfoot, M1 to 5 zone, hallux, and second to fifth toes (Figure 3B). The force applied to the M5 (N) was normalized by each subject's body weight (%BW). After calculating the variables during the movement using BodyBuilder and Footscan 9 software, Rstudio (version 4.1.1; The R Foundation for Statistical Computing) was used to perform time normalization with 100% of the time from initial contact to toe-off. IC was defined as the point at which the ground reaction force vertical component was >10 N, and toe-off was defined as the point at which the floor reaction force vertical component was <10 N. After these processes were performed on data from the 5 trials for each subject, the peak force to the M5, HIR angle (dynamic HIR), thigh ER angle, lower leg ER angle, ankle eversion angle, and FPA at the IC were calculated. All angle data excluding FPA were calculated at the time of peak force to the M5. For each variable, the mean value of the 5 successful trials was calculated and used in the statistical analysis.

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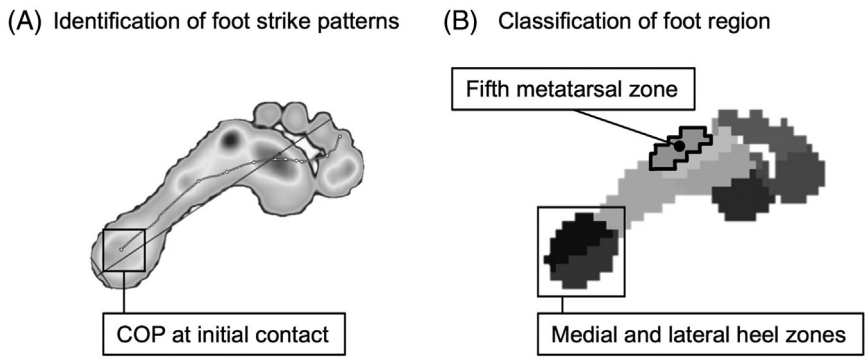

Figure 3 — Identification of foot strike pattern by Footscan and classification of foot region. Identification of foot strike patterns using Footscan (A) and classification of foot regions (B). The foot region was automatically classified by Footscan 9 software. The force applied to the M5 zone (B) was used as the load index for M5. COP indicates center of pressure; M5, fifth metatarsal.

Statistical Analysis

Shapiro–Wilk tests were performed for the participants' force to the M5, passive HIR, dynamic HIR, thigh ER angle, lower leg ER angle, ankle eversion angle, and FPA. As all variables followed a normal distribution, Pearson correlation coefficient was used to examine the relationship between parameters. Statistical analysis was performed using the statistical analysis software RStudio, with the significance level set at 5%.

Results

The foot strike patterns showed that the center of pressure at IC was distributed to the medial or lateral heel in all subjects (Figure 3B) and that the subjects had rearfoot contact. The trajectories of each parameter for all subjects and the related values (mean [SD]) are shown in Figure 4 and Table 1, respectively. The correlations between variables are shown in Table 2. There were significant positive correlations between passive HIR and dynamic HIR ($r = .50$, $P < .05$), between thigh ER and lower leg ER ($r = .66$, $P < .01$), between thigh ER and FPA ($r = .73$, $P < .01$), and between lower leg ER and FPA ($r = .88$, $P < .01$). Significant negative correlations were found between thigh ER and the force to the M5 ($r = -.54$, $P < .05$), between lower leg ER and the force to the M5 ($r = -.56$, $P < .01$), between FPA and the force to the M5 ($r = -.56$, $P < .01$), between lower leg ER and ankle eversion ($r = -.62$, $P < .01$), and between ankle eversion and FPA ($r = -.65$, $P < .01$).

Discussion

This study showed a significant positive correlation between passive HIR and dynamic HIR, but no significant correlation between HIR and the force to the M5; thus, part of the first hypothesis was not supported. A negative correlation was found between FPA and the force to the M5, which supports the second hypothesis. To our knowledge, no previous studies have examined the relationship among passive HIR, lower limb joint angles, and the forces on the M5 during soccer movements.

Regarding the foot strike pattern, all subjects exhibited a rearfoot strike pattern. In this study, the run-up was not a small step but a single step onto the Footscan. Therefore, the subjects needed to brake against greater forward shift of the center of gravity, which may have been a strategy to adapt to the environment of the movement task. The movement speed was similar to previous studies that had been conducted in the laboratory for change of direction[25,26] and 1 study conducted in a nonlaboratory environment.[27]

In a previous study, limitation of passive HIR could increase the force on the M5 by altering the kinematic chain during movement, which could be a risk for a Jones fracture.[13] In the present study, despite the positive correlation between the passive and dynamic HIR, there was no significant correlation between both passive and dynamic HIR and thigh ER, lower leg ER, or force on the M5. The observation might be due to that the dynamic HIR (5.8° [7.5°]) was much smaller than the passive HIR (39.9° [7.9°]); thus, no compensatory thigh ER due to HIR restriction appeared. These results suggest that the limitation of HIR is unlikely to be involved in Jones fracture in crossover cutting in the 45° direction, which is frequently performed during soccer competitions.[11,22]

Moreover, in this study, a greater FPA at IC was associated with greater lower leg ER and thigh ER. These results suggest that an increase in FPA at IC altered the movement of the proximal segment. Nevertheless, no significant correlation was found between these parameters and dynamic HIR, suggesting that the dynamic HIR magnitude changes with the magnitude of horizontal pelvic rotation because the supporting leg is fixed on the ground during the stance phase. However, orienting the FPA in a new direction from the IC can cause a greater thigh ER, and therefore, the angular momentum of the HIR induced by the horizontal pelvic rotation can be increased. As a result, the compensatory kinematic chain due to the HIR limitation and the increased force on the M5 could be prevented.

The correlation between FPA and ankle eversion was opposite to our hypothesis. Increased subtalar pronation due to increased FPA, as explained in a previous study,[15] was based on running analysis. The increase in FPA during running means that the longitudinal axis of the foot moves away from the line of progression, resulting in an increase in subtalar pronation and a decrease in ankle dorsiflexion.[15] This kinematic change is considered to be the result of a change in the rotational axis of the ankle relative to the direction of running due to an increase in FPA. In crossover cutting, the increase in FPA means that the ankle axis moves closer to the new line of progression, which may result in an increase in sagittal plane motion and a decrease in ankle eversion, in contrast to running.

Although a significant negative correlation was found between FPA at IC and the force to the M5, no significant correlation was

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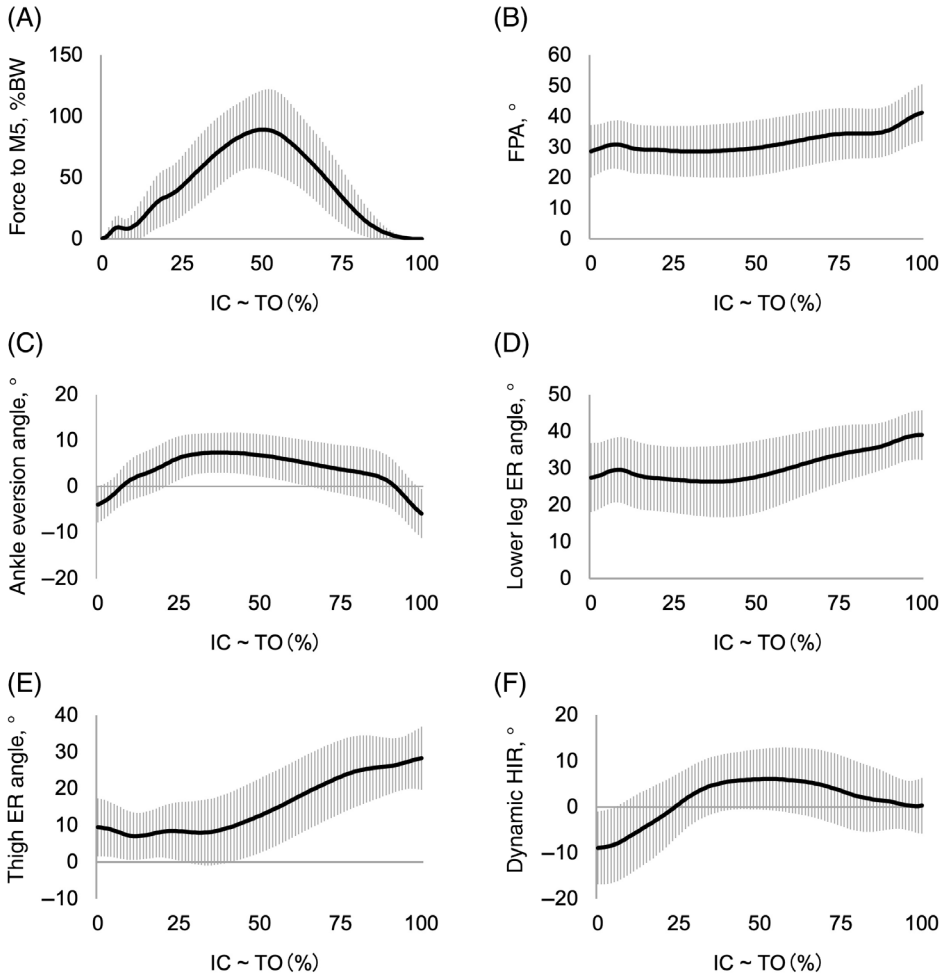

Figure 4 — Trajectory of each parameter during crossover cutting. IC to TO was normalized to 100%, and the black and gray lines indicate the means and SDs of all subjects, respectively. BW indicates body weight; ER, external rotation; FPA, foot progression angle; ER indicates external rotation; HIR, hip internal rotation; IC, initial contact; M5, fifth metatarsal; TO indicates toe-off.

Table 1 Average of Each Parameter for All Subjects

	Mean (SD)
Time of peak force to M5, %stance	51.3 (10.3)
Force to M5, %BW	105.5 (35.4)
Passive HIR, °	39.9 (7.9)
Dynamic HIR, °	5.8 (7.5)
Thigh ER, °	12.9 (9.2)
Lower leg ER, °	28.0 (9.24)
Ankle eversion, °	6.8 (4.7)
FPA at IC, °	29.7 (8.9)

Abbreviations: BW, body weight; HIR, hip internal rotation; ER, external rotation; FPA, foot progression angle; IC, initial contact; M5, fifth metatarsal.

found between ankle eversion and the force to the M5, partially rejecting the hypothesis. The effect of FPA on the force to the M5 may be due to changes in kinetics. Lai et al[17] investigated the relationship between FPA and ankle moment in walking[28,29] and between FPA and plantar pressure[16] and showed that an increase in FPA leads to greater ankle eversion moment and, consequently, greater pressure on the medial plantar surface of the foot. Since crossover cutting is a movement that exerts more pressure on the lateral plantar surface of the foot compared with running,[9] an ankle inversion moment can occur during the movement. Therefore, the increasing FPA in crossover cutting can decrease the ankle inversion moment, resulting in a decrease in the force to the M5. In this study, the ankle eversion angle decreased with increasing FPA, which may also have led to a decrease in the ankle inversion moment arm, supporting the conclusion.

The limitations of this study include the following. First, the causal relationship could not be examined. As this was a cross-sectional study, it is unknown whether the dynamic and passive HIR and FPA were causally related to the development of a Jones fracture. In the future, it will be necessary to clarify which movements are associated with a Jones fracture through prospective studies. Second, the force on the M5 was automatically classified by the Footscan software as a load index for the M5 zone. Jones fractures are considered to be caused by an increase in the bending moment of the proximal M5 due to a large force applied to the distal M5.[6] In the future, it is necessary to use methods to analyze the force applied to the distal M5 to clarify the detailed relationship with the angular parameters. Third, the angle of directional change during the movement task was set to 45°. Based on the results of this study, the dynamic HIR was lower

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Table 2 Correlations Between Variables

Parameters	Force to M5	Passive HIR	Dynamic HIR	Thigh ER	Lower leg ER	Ankle eversion	FPA at IC
Force to M5	1	−.04	.14	−.54*	−.56**	.18	−.56**
Passive HIR		1	.50*	.04	−.04	−.17	−.03
Dynamic HIR			1	−.28	−.41	−.07	−.18
Thigh ER				1	.66**	−.34	.73**
Lower leg ER					1	−.62**	.88**
Ankle eversion						1	−.65**
FPA at IC							1

Abbreviations: BW, body weight; HIR, hip internal rotation; ER, external rotation; FPA, foot progression angle; IC, initial contact; M5, fifth metatarsal.
*$P < .05$. **$P < .01$.

than the passive HIR during crossover cutting in the 45° direction. In this study, the angle of directional change was set to 45° based on the angle frequently used during soccer games.[11,22] However, in a study on cutting maneuvers, kinematics and kinetics vary depending on the angle of change of direction.[26] Considering the use of HIR in crossover cutting,[10] it is possible that the greater the angle of change in direction, the greater the HIR required during movement. In the future, it will be necessary to verify the HIR for movements that involve angles >45°, which require a greater HIR during movement. Fourth, this study could not be conducted in an environment that simulates an actual competitive situation. In this study, participants performed the movement task barefoot, and the ground contact surface of the foot differed from that of the ground during soccer games. Environmental factors that may be associated with the risk of a Jones fracture include the type of cleats[30] and the nature of the playing surface.[31,32] Therefore, setting the experimental environment to resemble actual playing conditions closely and examining kinematic factors related to the forces acting on the M5 under these conditions can lead to a more high-quality study.

In this study, the HIR in the passive and dynamic conditions were not related to the force applied to the M5 during crossover cutting. However, the movement method that aligns the FPA with the new direction was related to a reduction in the force on the M5. Therefore, instructing athletes in movements that increase FPA from IC could help prevent a Jones fracture.

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